

Prangan Nandy

prangan.nandy.8@gmail.com | [linkedin.com/in/prangan-nandy/](https://www.linkedin.com/in/prangan-nandy/)

EDUCATION

University of South Carolina

Master of Science in Mechanical Engineering – GPA: 4.00/4.00

Columbia, SC, USA

June 2026 (Expected)

Bangladesh University of Engineering and Technology

Bachelor of Science in Mechanical Engineering – GPA: 3.60/4.00

Dhaka, Bangladesh

May 2022

SKILLS

Simulation & Modeling: COMSOL, ANSYS, STAR-CCM+, FactSage

CAD: SolidWorks, AutoCAD

Experimental: ICCD Imaging, Spectroscopy, Laser Diagnostics

Programming Languages: Python, MATLAB

EXPERIENCE

RASAER LAB, University of South Carolina

Columbia, SC, USA

Graduate Research Assistant

Jan 2024 - Present

- Developed 2D multiphysics model to simulate non-equilibrium plasma generation using piezoelectric transformers, predicting breakdown behavior and explaining surface discharge mechanisms
- Conducted experimental diagnostics using ICCD imaging and optical emission spectroscopy, demonstrating stable plasma discharge at low power input (~2 W)
- Designed and fabricated custom laboratory systems including gas chambers (helium plasma reactor) and vapor delivery setups (to identify effect of water on plasma) for controlled experimentation
- Investigated plasma-assisted conversion of landfill gases by using a global chemical kinetics model which reproduced experimentally observed oligomer growth trends

INDETECH SOFTWARES LIMITED

Dhaka, Bangladesh

Assistant Engineer

March 2023 – Dec 2023

- Built a Python-based simulation model to analyze cooling plant water consumption, reducing water requirement by 10% through operational optimization
- Created standardized templates for Environmental Product Declarations (EPD) and Safety Data Sheets (SDS) for integration into a material passport database

ABUL KHAIR STEEL PRODUCTS LIMITED

Chittagong, Bangladesh

Technical Management Trainee

July 2022 – Feb 2023

- Conducted root cause analysis of mechanical failures of cold rolling mill, improving system reliability and operational stability
- Designed a roll-changing mechanism using SolidWorks, reducing maintenance time and increasing operational profit by \$9000 per week
- Developed P&ID diagrams using AutoCAD for steam delivery systems to support plant operations and maintenance

Certification

SolidWorks Associate – Mechanical Design